Population and Migration

Based on:

PRLB chapter 7

Barro and Sala-i-Martin (2004)

Clemens (2011) + (2022).

Monday 23 February 2026

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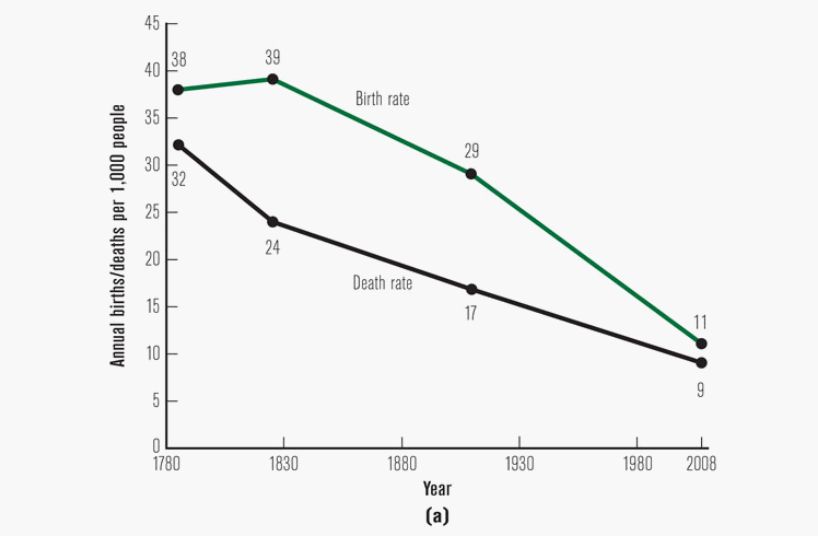
Population Growth

Motivation

- Robert McNamara (70ties): "The greatest single obstacle to the economic and social advancement of the majority of the peoples in the underdeveloped world is rampant population growth"
- James Wolfensohn (90ties): Limited focus on population growth.
- Who is right?

History

- Population growth close zero until around 1800.
- Introduction of settled agricultural revolution changed the picture despite famines, plagues, and wars.
- Industrial Revolution.
- Annual population growth 1800-1945: 0.6 percent
- Demographic transition



First stage: Both high birth and death rates.

Second stage: Death rates declining faster than birth rates.

Third stage: Birth rates declining faster than death rates.

Fourth stage: Zero population growth (again)

TODAY: Almost all population growth in high-income countries is attributable to immigration.

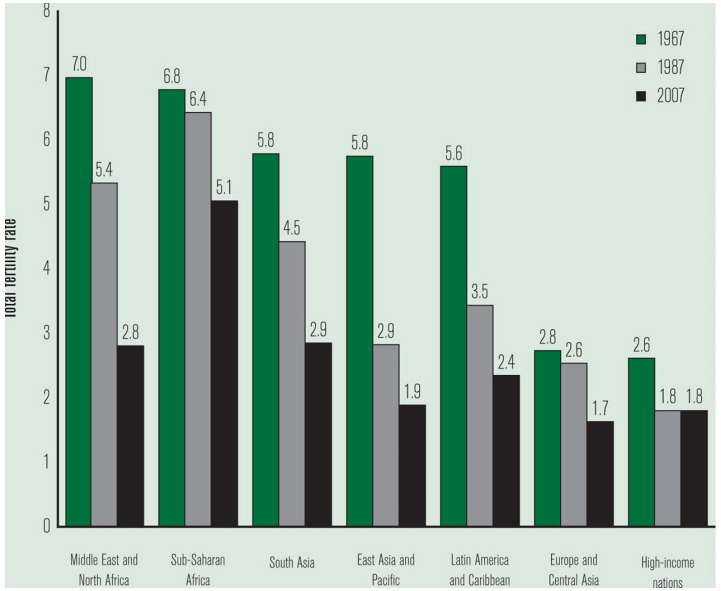
Future

- Even today considerable variation in fertility across regions (and countries) remain.
- Will the total fertility rates (TFR) stay above replacement rates?
- Three scenarios and three reasons:

(1) Desire for large families

(2) Unwanted children hypothesis

(3) Population momentum



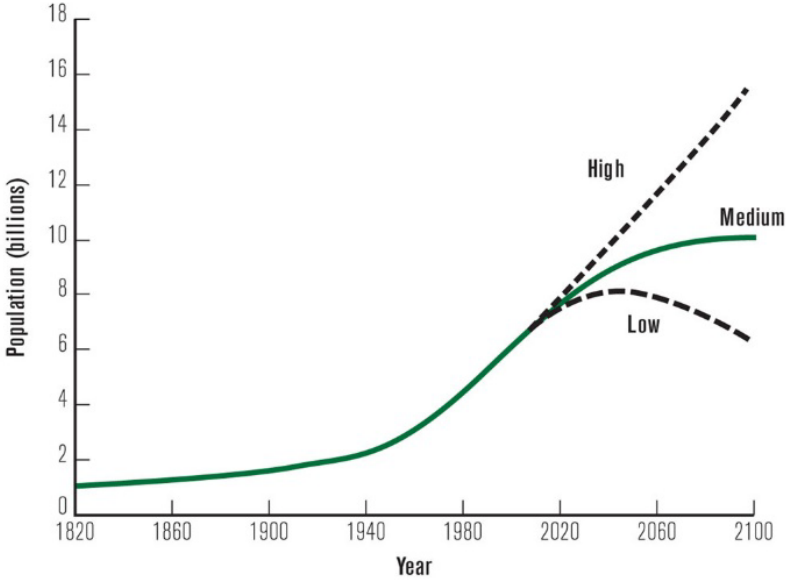


FIGURE 7-4 World Population Historical Trends and Projections

UN population projections for 2000–2100. The scenarios presented here represent different future levels of fertility.

Source: Angus Maddison, "Statistics on World Population, GDP, and Per Capita GDP, 1–2008 AD," www.ggdc.net/MADDISON/Historical_Statistics/vertical-file_02-2010.xls. U.N. Department of Economic and Social Affairs, Population Division, Population Estimates and Projection Section, *World Population Prospects, The 2010 Revision*, October 20, 2011, available at http://esa.un.org/unpd/wpp/index.htm; accessed February 2012.

Causes

- How is “n” determined and why has it declined?
  - Perhaps the negative correlation means that fertility declines with economic development?
- What is the impact of population growth on per capita income?
  - Is the negative correlation between “n” and “Y/L” due to a detrimental impact from population expansion on prosperity?
- Depending on answers: Where does this leave us in terms of policy options?

How is n determined

- Until the end of the 18th century: periods of plenty associated with greater family size everywhere
  - Pattern consistent with the Malthusian model (familiar from economic history): “The passion between the sexes would cause population to expand as long and far as food supplies permitted”.

- Subsequently, fertility patterns change fundamentally
  - Rising prosperity associated with smaller, not bigger, families.

- The reversal is associated with the demographic transition.
  - Proto-typical demographic transition (first mortality transition and then fertility transition).

How is n determined

Determinants of mortality

What caused the mortality transition?

- Nutritional improvements.
- Improvements in sanitation.
- Technology (especially, Germ theory early on; later instances also the international epidemiological transition associated with penicillin).

The relative importance of each of these elements is in debate: McKeown and Fogel (nutrition) vs Preston (tech).

How is n determined

Determinants of fertility – Traditional population pessimists

- Lower fertility – less investments needed to provide a constant amount of capital per worker (**capital widening**) or investments used to increase capital per worker (**capital deepening**).

- Lower fertility - **Public funds** can be **redirected** away from health and education toward physical capital.

- Lower fertility – slower population growth – lower **dependency ratio** – reduced consumption and increased savings (at given income level) – higher capital accumulation. (**demographic dividend**)

How is n determined

Determinants of fertility

- What caused the fertility transition?
  - Range of theories exist.
- Income and substitution effects.
  - Assume children associated with time cost. At low levels of income, the income effect dominates; at high levels of income the substitution effect dominates.
- Mortality.
  - Precautionary motive for having kids (limiting the risk of being childless). Declining mortality reduces desired family size. Countervailing: Cost of non-surviving kids.
- Quantity/Quality.
  - Accelerating technological change increases the return to skill investments. Provides an incentive to reduce family size yet expand investments in each child (substituting “quantity” for “quality”).
- Reinforcing mechanisms: Norms, Gender roles, Child labor.

How is n determined

Word of caution

- Demographers usually favor mortality explanations, as the mortality transition precedes the fertility transition.
  - Imagine that the speed of technological change accelerates. Implications (in theory):
    - (i) Increases income per capita, therefore improves nutrition and lowers mortality.
    - (ii) Makes more investments per child (e.g., schooling) more attractive and thus lowers fertility if Q-Q trade-off is operative.
- BUT: Its very likely that (i) simply sets in faster than (ii).

- **Conclusion**: In theory, the relative timing of the two transitions tells us little (if anything) about mechanisms.

Impact of n on y

Population pessimists

- The most basic reason why population growth may reduce prosperity is **Capital Dilution** (population pessimists)
- Assume standard Solow model with CD production function such that $y = A^{1-\alpha}k^{\alpha}$. Assessed in steady state:

$$y^* = A\left[\left(\frac{K}{Y}\right)^*\right]^{\frac{\alpha}{1-\alpha}} = A\left[\frac{s}{n+\delta}\right]^{\frac{\alpha}{1-\alpha}}$$

- Assume standard assumptions ($\alpha = 1/3$ and δ=0.05) yields a modest impact (country 1: n=1% country 2: n=3%)

$$\frac{y_1}{y_2} = \left(\frac{0.03 + 0.05}{0.01 + 0.05}\right)^{1/2} = 1.15$$

- Modest income per capita differences
- Key reasons why population growth may lower prosperity
  - Limits physical capital per capita (**capital dilution**) and thus labor productivity.
  - Limits natural resources per capita (fixed factor) and thus labor productivity (extended Solow model).

Impact of n on y

Population optimists

Reasons why a bigger population (or population growth) may be good for growth:

1. Division of labor (Adam Smith)
2. Technology adoption in the presence of population pressure (Ester Boserup)
3. More people, more ideas / non-rivalry of ideas (Phelps-Jones)

Accordingly, it is an empirical matter which theories seem to dominate.

Impact of n on y

Empirics

Mankiw-Romer-Weil (1992) demonstrate a negative link (correlation) between n and y?

Problems:

- n = f – d (f=fertility, d= death). Slower n could be caused by higher d as well as lower f.
- Fertility influenced by income (simultaneity bias)
- Being poor can also lead to population growth (Q-Q theory) (omitted variable bias)

Several papers have tried to disentangle exactly these endogeneity problems (advanced development economics courses look deeper into this).

International Migration - Model

Migration citations

- "From 1990 to 2014, U.S. economic growth would have been 15 percentage points lower without the benefit of migration". Citi Research (2018).
- "There's a way for President Trump to boost the economy by four percent, but he probably won't like it…For every 1 percent increase in U.S. population made of immigrants, GDP rises 1.15 percent. So, a simple way to get to Trump's 4 percent GDP bump? Take in about 8 million net immigrants per year". ProPublica (2017)
- However, few academic studies document the direct link between emigration/immigration and growth/development (Thesis!!!).

Solow model with migration

- Let M, which can be positive or negative, be the flow of migrants into a domestic economy and κ the quantity of capital that each migrant brings along.
- The domestic population and labour force, L, grow due to fertility net of mortality at the constant, exogenous rate n. The overall growth rate is therefore:

$$\frac{\dot{L}}{L} = n + \frac{M}{L} = n + m$$

- The change in capital stock is given by

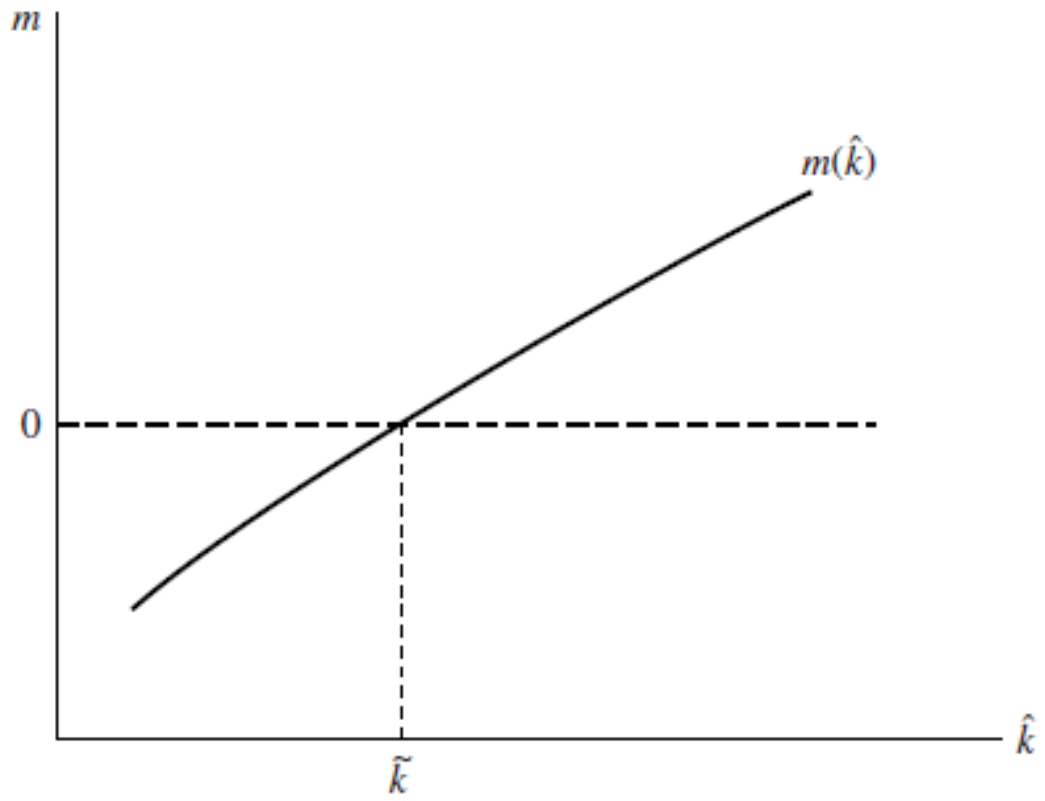
$$\dot{K} = sF(K, L) - \delta K + \kappa M$$

- **The new element is that $\boldsymbol{\kappa M}$** (the capital brought in by immigrants or taken out by emigrants) **contributes to capital accumulation**.
- The growth rate can be derived as:

$$\frac{\dot{\hat{k}}}{\hat{k}} = \frac{sf(\hat{k})}{\hat{k}} - (x + n + \delta) - m(1 - \frac{\hat{\kappa}}{\hat{k}})$$

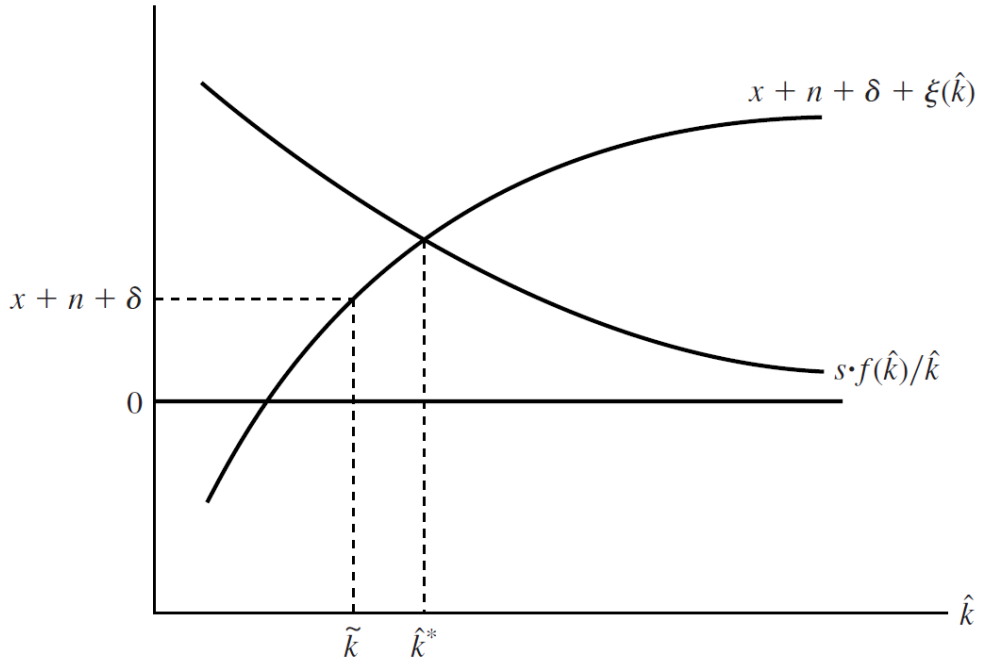
where $\hat{\kappa} = \kappa e^{-xt}$ is the capital per effective immigrant.

The Migration Curve



- **The migration rate.** For given conditions in other economies, a higher value of *k* raises the domestic wage rate and tends accordingly to increase the net migration rate, *m*. (The interaction between the migration curve and the horizontal curve is the quantity of capital per effective worker that yields a zero net migration rate).
- A reduction of wage rates in foreign countries makes migration to the domestic country more attractive and shifts the function *m(k)* upward.
- The slope of the function depends on the relation between the cost of moving (for the marginal migrant) and the volume of migration.
- If this cost increases rapidly with the number of migrants, then a change in *k* has only a small effect on migration; that is, the curve *m(k)* is relatively flat.

Growth diagram



- If $\hat{k} > \tilde{k}$ then $m(\hat{k}) > 0$ and vice versa. We assume the intersection such that $\hat{k}^* > \tilde{k}$,$m^* > 0$ and the domestic economy is a recipient of migrants in steady state.
- *ξ(k)* is the negative effect of the migrants' human capital on the growth rate of the domestic capital stock.

What if conditions worsen in other economies and there is an immigration pressure?

Then m(k) would shift upwards (and so will the curve effective depreciation). k* falls and m* rises.

An expansion of the supply of immigrants lowers the steady state capital intensity in the domestic economy.

This result follows because the immigrants come with relatively little capital.

Note: Transitional dynamics and convergence will depend on the sensitivity of migration to k.

Evidence: Increase in convergence speed across countries due to migration. However, differences between sending and receiving countries (see Clemens 2011 and 2022).

International Migration - Empirics

Definitions

- **Emigration** is the act of leaving a resident (host) country with the intent to "permanently" settle elsewhere. Host country perspective.
- **Immigration** is the international movement of people to a receiving (destination) country of which they are not natives or where they do not possess citizenship. Destination country perspective.
- A migrant *emigrates* from their country of origin, and *immigrates* to their new choice of origin.
  - Thus, both emigration and immigration describe **migration**, but from different countries' perspectives.

Estimated gains from eliminating labour mobility barriers

Efficiency Gain from Elimination of International Barriers
(percent of world GDP)

All policy barriers to merchandise trade	
1.8	Goldin, Knudsen, and van der Mensbrugghe (1993)
4.1	Dessus, Fukasaku, and Safadi (1999)[a]
0.9	Anderson, Francois, Hertel, Hoekman, and Martin (2000)
1.2	World Bank (2001)
2.8	World Bank (2001)[a]
0.7	Anderson and Martin (2005)
0.3	Hertel and Keeney (2006, table 2.9)
All barriers to capital flows	
1.7	Gourinchas and Jeanne (2006)[b]
0.1	Caselli and Feyrer (2007)
All barriers to labor mobility	
147.3	Hamilton and Whalley (1984, table 4, row 2)[c]
96.5	Moses and Letnes (2004, table 5, row 4)[c]
67	Iregui (2005, table 10.3)[c,d]
122	Klein and Ventura (2007, table 3)[e]

- More than 40 percent of adults in the poorest quartile of countries "would like to move permanently to another country" if they had the opportunity.
- **How large are the economic losses caused by barriers to <u>emigration</u>?**
  - Main question in Clemens (2011)
- NB!!! The gains to eliminating barriers to <u>emigration</u> is 1-2 times larger than the gains from dropping all remaining restrictions on international flows of goods and capital.

Estimated gains from eliminating labour mobility barriers

Efficiency Gain from Partial Elimination of Barriers to Labor Mobility

	Removal of barriers	*Net emigration rate (% origin-region population)*	*Efficiency gain (% world GDP)*
Moses and Letnes (2004, 2005)	Complete	73.6	96.5
	Partial	29.3	54.8
	Partial	10.3	22.0
Iregui (2005)	Complete	53	67
	Partial	24	31
Klein and Ventura (2007)	Complete	>99	122
	Partial	14.8	20
	Partial	7.3	10
Walmsley and Winters (2005)	Partial	0.8	0.6
	Partial	1.6	1.2
van der Mensbrugghe and Roland-Holst (2009)	Partial	0.8	0.9
	Partial	2.0	2.3

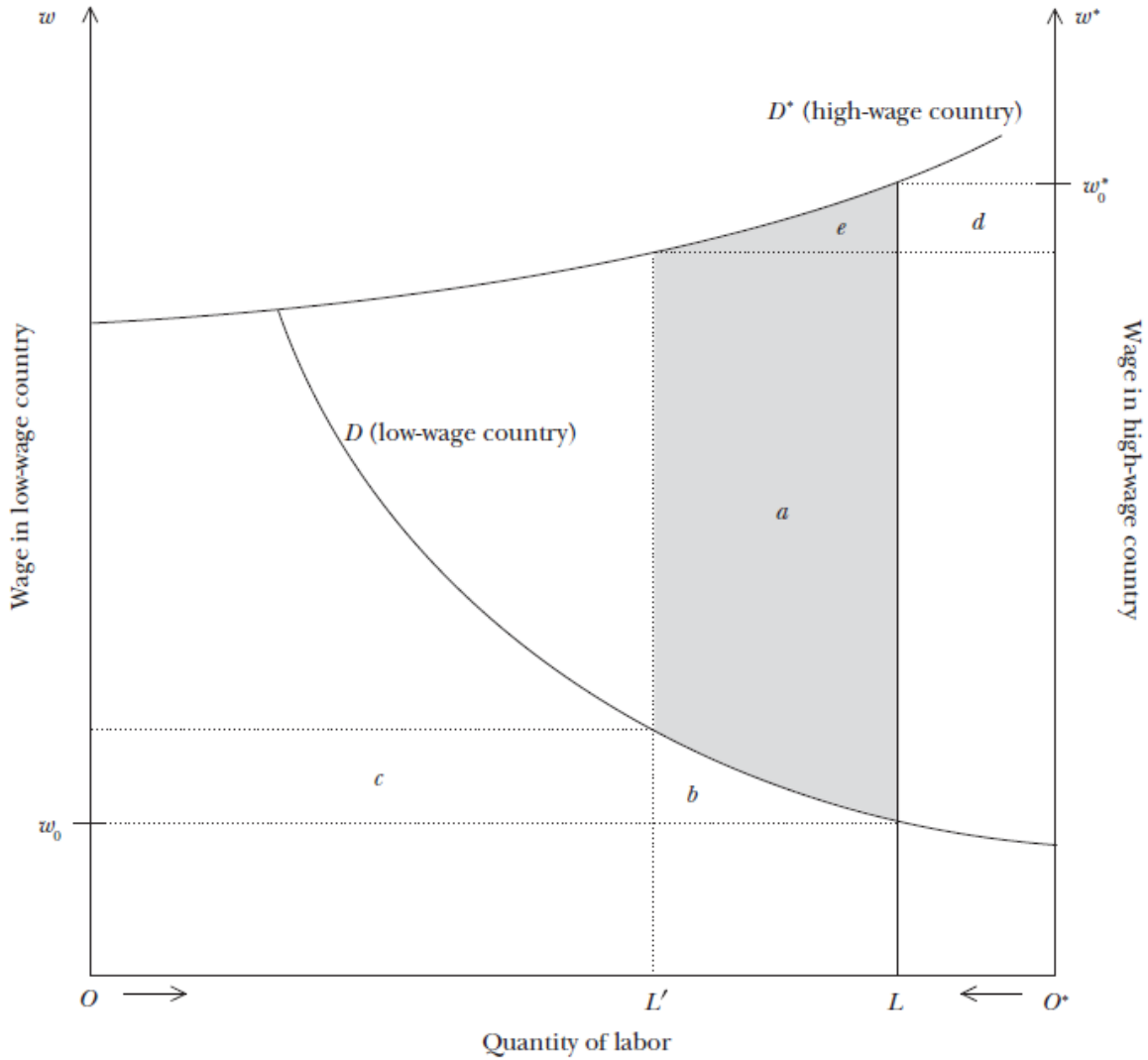
Estimates suggest that the gains from reducing emigration barriers are enormous.

Tens of **trillions** of dollars to gain - much more than development aid.

The four big questions

1) What are the external effects of (especially skilled) emigrants' departure on the productivity of non-emigrants? (Brain drain)
   - Many of the above estimates rest on the assumption that this effect is small or nil.
2) What is the elasticity of labor demand, in the origin and destination countries?
   - Are the above studies getting it right?
3) How much of international differences in productivity depend on workers' inherent **traits**—accompanying them when they move—and how much depends on their **surroundings**?
   - Is productivity mostly about who you are, or where you are?
4) Given the many barriers that prevent emigration today, what future level of emigration is feasible?

Determinants of the gains from Emigration



- **Migrants** gain welfare corresponding to area *a* + *b*.
- **Low-income country**:
  - Labor gains area *c*.
  - Owners of other factors lose area *b* + *c*.
- **High-income country**:
  - Labor loses area *d*.
  - Owners of other factors gain area *d* + *e*.
- **Total global welfare**:
  - Gain is *a* + *e* (the shaded area).

Relationship to the four big questions:

- Emigration externalities on non-migrants, could lead to downward shifts in demand curves, offsetting positive gains.
- If labor demand becomes highly elastic at the **origin** or at the **destination**, the gain shrinks.
- If emigrants are inherently less productive than non-migrants at the destination, the true demand curve for their labor lies further below D*, and the gain also shrinks.
- As the size of feasible migration shrinks, L' gets closer to L, and once again, the gain shrinks.

Evidence of the "big four"

- Little evidence to support the notion that externalities from labor mobility would greatly affect the global welfare estimates presented above.
- Estimated elasticities in various research show that the estimates used for the global welfare calculation are consistent:
  - A 10% removal of emigration barriers leads to a 3–4 percent increase in wages for non-emigrants at the **origin**, and a 2.5 percent decline at the **destination**.
  - Substantial adjustment of elasticities is unlikely to alter the estimated efficiency gains.
- Can better accounting for unobserved differences in the determinants of productivity between migrants and non-migrants greatly alter the estimates of efficiency gains from greater labor mobility? **Clemens: NO.**
- Substantial expansion of emigration "politically impractical".
  - But: Global efficiency gains from even small relaxations of existing barriers to emigration are large.

Clemens conclusion

"The available evidence suggest that the gains to lowering barriers to emigration appear much larger than gains from further reductions in barriers to goods trade or capital flows—and may be much larger than those available through any other shift in a single class of global economic policy"

Empirics versus Models

- Traditional migration model:
  - Increase in global migration driven by (an effect of) poverty in migrant-origin countries. Push factor.
  - Increase in global migration will result in (be the cause of) poverty in destination countries (if low-skill) or origin countries (if high-skill).
- Questions to ask (and evidence):
  - Did migration arise from failed economic development in poor countries?
    - Just the opposite. Mexicans are not in the US because Mexico fell into poverty, but because Mexico emerged from poverty.
  - Did more migration result from a retreat of trade and capital flows?
    - Just the opposite: They soared during the same years.
  - Did initial waves of migration reduce the incentive for further migration by spatially equilibrating the labor market?
    - Just the opposite.

Effects of Rising Global Mobility

- Did the large rise in migration from the developing world substitute for workers in the destinations? **NO**
  - But it did course wage pressures in origin countries.
  - + remittances higher than aid and causing lasting increases in human capital improvements.
- Did migration typically select for the least productive workers? **NO**
- Did more migration substitute for other forms of globalization via factor-price equalization? **NO**
  - Global wealth increased and modestly reduced (between country) inequality as a result of migration.
  - But moderate increases in (within-country) inequality at the destinations as a result.
- Our standard migration models are not capturing this very well.